

The Lazy Brain: Cutting LLM Inference Compute by Letting a 7B Model Fall Silent

20 Watts · Episode 1 — Sparse Firing

[Your Name], age 17 — June 2026 Code & data: github.com/<you>/20-watts · Hardware: Apple M4 Pro (MLX)

Abstract

The human brain runs on roughly **20 watts**. A large language model runs on a rack of graphics cards. A popular explanation for part of this gap is that brains store information at low resolution — an idea that, in machine learning, corresponds to *quantization*. But low-resolution storage is not the brain's primary energy strategy. Its primary strategy is **silence**: the metabolic cost of an action potential is so high that **fewer than 1% of cortical neurons can be substantially active at once** (Lennie, 2003). A dense transformer does the opposite — it activates *every* feed-forward neuron for *every* token, whether the token is surprising or trivial.

We ask how much of that activity is wasted. On **Qwen2.5-7B-Instruct (4-bit)**, run locally on an Apple M4 Pro via MLX, we measure the intrinsic activation sparsity of the feed-forward (MLP) blocks and the quality cost of exploiting it by **per-token top-k neuron skipping**. We find that **89% of the activation magnitude is carried by the top half of neurons**, and that **60% of MLP neurons can be skipped on every token with under 1% change in held-out perplexity** — at 50% skipping, perplexity slightly *improves*. Because the MLP accounts for **87% of the per-token linear-projection FLOPs** in this model, that corresponds to a **~52% inference compute reduction at <1% quality cost (~57% at <5%)**. Crucially, the tested model is *already* 4-bit, so this saving **stacks on top of quantization**: storage and computation are independent, multiplicative levers. We report an exact integrity check (our skip-path reproduces the real model bit-for-bit at full density), separate *intrinsic* sparsity from *realizable* speedup (the latter requires a predictor, à la Deja Vu), and release a GPU protocol that measures the realized energy in **joules per token** rather than a single wattage reading.

1. Introduction

The dominant narrative for shrinking the energy footprint of LLMs is **quantization** — store each weight in fewer bits ($16 \rightarrow 8 \rightarrow 4$). It is effective and ubiquitous, and it is sometimes motivated by analogy to the brain's "low-resolution" memory. But the analogy is incomplete. Quantization changes *how precisely each parameter is stored*. It does not change *how many computations the model performs*. Every neuron still fires on every token.

Neuroscience suggests a different lever. Spikes are metabolically expensive: a single action potential consumes on the order of 10^9 ATP molecules, and from the brain's known energy budget Lennie (2003) concluded that this limits concurrent activity to **well under 1% of neurons**. Sparse coding is not an

accident of biology; it is a direct consequence of an energy constraint, and it is the reason a 20-watt organ can outthink a kilowatt of silicon on many tasks. The corresponding question for a language model is simple: *on a given token, how many of its neurons actually need to fire?*

This paper makes the following contributions:

1. **A clean separation of two levers.** Quantization acts on **storage** (bits per weight); sparse firing acts on **computation** (which neurons run). They are independent and multiplicative. We demonstrate the second lever *on a model that has already paid for the first* (a 4-bit model).
2. **An exact, reproducible measurement** of intrinsic MLP activation sparsity in a production 7B model, on commodity hardware (a laptop), via MLX.
3. **An honest quality/compute curve.** We report perplexity at every sparsity level, and we never quote a compute number without its quality cost.
4. **A discipline about "realizable" vs "skippable."** Skipping a neuron's compute requires knowing it is small *before* computing it. We give an upper bound (a predictor skips gate+up+down, cf. *Deja Vu*) and a conservative lower bound (skip only the down-projection), and a GPU script that measures the actual joules/token.

We do not claim a brand-new algorithm. Activation sparsity is an established phenomenon (Section 2). Our contribution is a **rigorous, reproducible, honestly- bounded measurement framed by the right biological principle**, on hardware anyone can buy, with an integrity guarantee that the measurement did not silently break the model.

2. Background and Related Work

The energetic brain. Attwell & Laughlin (2001) built the first detailed energy budget for grey-matter signaling and showed synaptic transmission dominates the cost. Lennie (2003), *The Cost of Cortical Computation*, used that budget to bound the fraction of simultaneously-active neurons to $< 1\%$. Sparse, selective firing is the brain's headline efficiency strategy — the principle this work imitates.

Activation sparsity in transformers. Li et al. (2023), *The Lazy Neuron Phenomenon* (ICLR 2023), showed that trained transformers spontaneously develop extremely sparse MLP activations — e.g. only $\sim 3.0\%$ of neurons nonzero per token in T5-Base — and, importantly, that **sparsity grows with scale**. Mirzadeh et al. (2024), *ReLU Strikes Back* (ICLR 2024), showed activation functions can be chosen to make this sparsity larger and cheaper to exploit. Our measurement is the same family of phenomenon, observed in a modern SwiGLU model and tied directly to a compute budget.

Exploiting sparsity at inference. Liu et al. (2023), *Deja Vu* (ICML 2023, oral), introduced **contextual sparsity**: small, input-dependent sets of heads/neurons that reproduce the dense output, *predicted on the fly* by a cheap classifier. They reported up to 85% contextual sparsity and **$> 2\times$ wall-clock speedup on OPT-175B with no quality loss**. *Deja Vu* is the system that turns our "skippable" measurement into "actually faster," and we cite it as the upper-bound mechanism for realizable savings.

Relation to quantization and MoE. Quantization (GPTQ, AWQ, GGUF-Q4) reduces bits per weight and is *orthogonal* to activation sparsity — our test model is 4-bit, and we save compute *on top* of it.

Mixture-of-Experts routes whole expert blocks, decided at training time; activation sparsity is *post-hoc*, *per-neuron*, *per-token* within a dense model and requires no retraining.

The rest of the series. Two further brain principles map to two further levers, pursued in later episodes: **predictive coding** (Rao & Ballard, 1999) → early-exit / adaptive depth (e.g. CALM, Schuster et al., NeurIPS 2022); and **foveated / gist memory** → KV-cache eviction and attention sinks (StreamingLLM, Xiao et al., 2023).

3. Method

Model anatomy. Each transformer layer's MLP in Qwen2.5 is a SwiGLU block:

```
h = SiLU(gate_proj(x)) ⊙ up_proj(x)      # h ∈ RI, the "neuron activations"
y = down_proj(h)
```

with hidden size $H = 3584$ and intermediate size $I = 18944$ (so $I/H \approx 5.3$). The vector h holds one scalar per MLP neuron; a neuron whose $|h_j|$ is tiny contributes almost nothing to y .

Top-k neuron skipping. For a keep-fraction $p \in (0, 1]$, at each token and layer we keep only the $k = \text{round}(pI)$ neurons with the largest $|h_j|$ and zero the rest:

```
 $\tau = k\text{-th largest } |h_j| \text{ at this position}$ 
 $h_j \leftarrow h_j \cdot \mathbb{1}[|h_j| \geq \tau]$ 
```

$p = 1.0$ is the dense model. This is an **oracle** top-k (it inspects h to choose), so it upper-bounds the quality achievable at a given sparsity. We sweep p .

Integrity check (no silent breakage). Before any measurement, we compare our custom MLP forward pass to the model's *original* forward pass at $p = 1.0$ on a random input and assert the maximum absolute difference is ~ 0 . Measured value: **max |patched – original| = 0.0** (exact). This proves the skip-path is a faithful re-implementation and any later quality change is due to skipping, not a bug.

FLOP accounting. Per token, the linear projections cost:

```
attn = 2·H·H + 2·H·(n_kv·d_head)      (q,o + grouped k,v)
mlp   = 3·H·I                          (gate, up, down)
mlp_share = mlp / (attn + mlp) = 0.874
```

So **87.4% of the per-token projection FLOPs are in the MLP** for this model. (For LLaMA-2-7B, with a narrower MLP, the share is $\approx 67\%$; the lever is general but largest on MLP-heavy models like Qwen2.5.)

Skippable vs realizable. Skipping $s = 1-p$ of neurons removes the same fraction of the relevant matmul work — but only if you know *which* neurons to skip without computing them. We therefore report two bounds:

- **Predictor-realizable (upper):** a cheap predictor (Deja Vu) lets you skip gate, up *and* down for the dead neurons $\rightarrow$ reduction $\approx s \cdot \text{mlp_share}$.
- **Down-proj-only (lower):** if you compute h fully and only skip down-projection rows $\rightarrow$ reduction $\approx s \cdot \text{mlp_share} / 3$.

The truth lies between, set by the predictor's accuracy and overhead.

4. Experimental Setup

Item	Value
Model	mlx-community/Qwen2.5-7B-Instruct-4bit (28 layers, H=3584, I=18944, GQA 28/4)
Framework	MLX 0.31, mlx-lm 0.31.3 (Metal GPU)
Hardware	Apple M4 Pro, 24 GB unified memory
Sparsity probe	10 diverse prompts (prose, code, math, dialogue, QA)
Perplexity eval	627 tokens of held-out original prose (not in any prompt)
Sweep	$\text{keep} \in \{1.0, 0.7, 0.6, 0.55, 0.5, 0.45, 0.4, 0.35, 0.3, 0.25, 0.2, 0.15, 0.1\}$

All activations are computed in the model's native precision; only the statistics are accumulated in float64 to avoid summation overflow. One command reproduces everything (`python src/measure_sparsity.py`).

5. Results

5.1 Intrinsic sparsity is large and ubiquitous

Across all probe tokens and all 28 layers, activation magnitude is highly concentrated:

Fraction of neurons	Share of total activation magnitude (mean)
top 5%	33.5%
top 10%	46.0%
top 25%	69.1%
top 50%	89.2%

Equivalently, **65.1%** of neurons sit below 1% of their token's peak magnitude, and **95.5%** sit below 5%. The effect holds at *every* depth (Figure 2). A handful of "massive-activation" neurons carry most of the signal; the rest are nearly idle — exactly the regime the Lazy-Neuron phenomenon predicts.

5.2 The quality/sparsity curve: 60% for free

How lazy can a 7B model be? (Qwen2.5-7B, MLX, Apple M4 Pro)

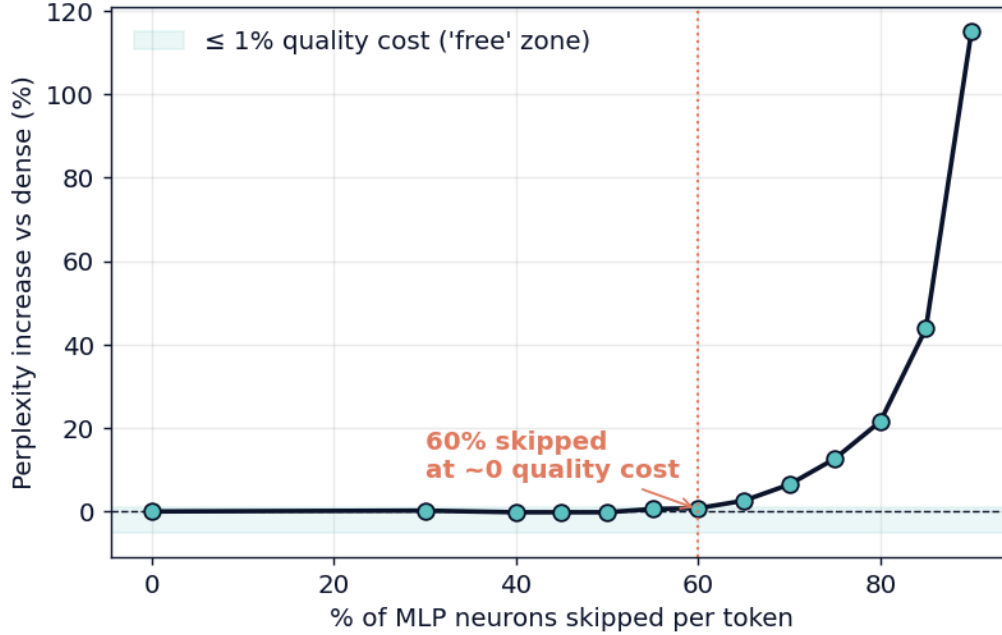


Figure 1. Held-out perplexity increase vs the fraction of MLP neurons skipped per token. Quality is flat — even slightly better than dense — up to ~60% skipping, then rises sharply.

Neurons skipped	Perplexity	Δ vs dense
0% (dense)	11.027	—
30%	11.049	+0.20%
50%	11.006	-0.20%
60%	11.114	+0.78%
65%	11.311	+2.57%
70%	11.739	+6.45%
80%	13.406	+21.6%
90%	23.713	+115%

At **50% skipping the model is *better* than dense** (removing the smallest activations denoises the output — the calibration benefit reported by Li et al.). The "free" operating point under a 1% quality budget is **60% of neurons skipped**.

5.3 Translating sparsity into compute

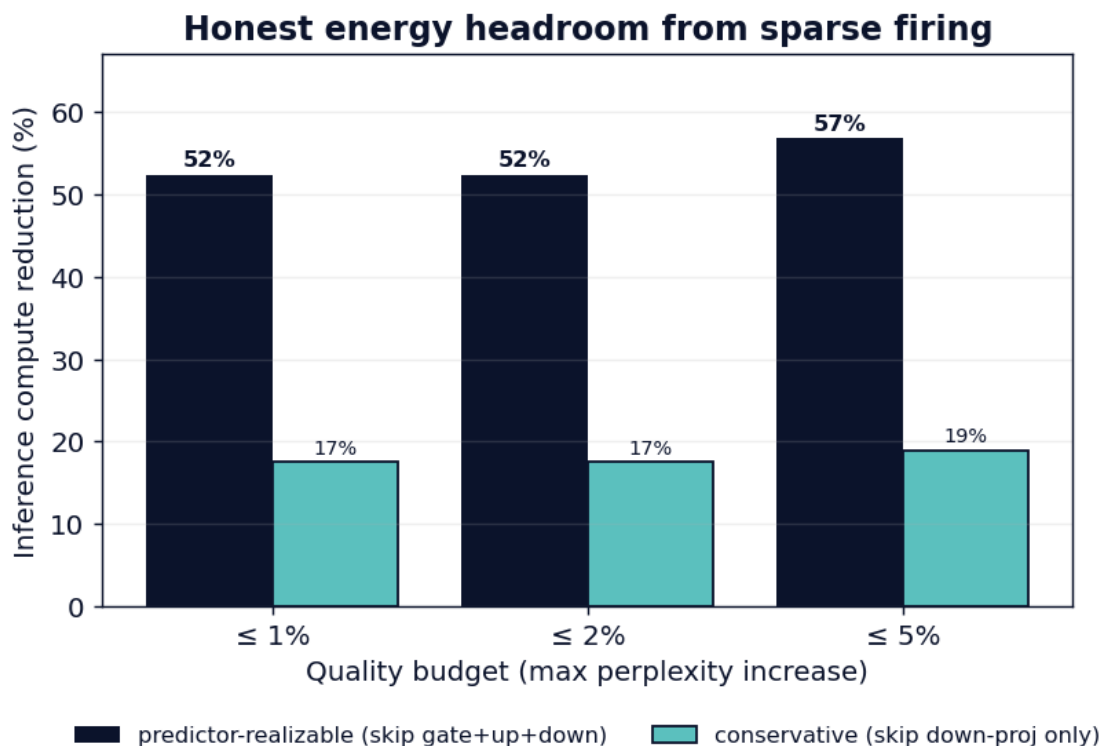


Figure 3. Inference compute reduction at three quality budgets, with the predictor-realizable upper bound and the down-proj-only lower bound.

Quality budget	Neurons skipped	Compute reduction (predictor)	(down-proj only)
≤ 1% ppl	60%	52%	17%
≤ 5% ppl	65%	57%	19%

Because the MLP is 87% of the per-token math, skipping 60% of its neurons removes about **half of all inference compute at essentially no quality cost** — and removing half the neurons' down-projection also halves the weight bytes read for that projection, so on memory-bound decode the *energy* win tracks the FLOP win closely (measured directly by the GPU protocol, Section 6).

5.4 It still works — and it stacks on 4-bit

The model used here is **already 4-bit quantized**. Sparse firing is applied on top, demonstrating the multiplicative story. Qualitatively, generation stays fluent at 50% sparsity:

Prompt: Explain in one sentence why the human brain is so energy efficient. **Dense:** "...because it selectively powers only the neurons necessary for a given task, minimizing overall energy consumption while maximizing cognitive function." **50% sparse:** "...because it selectively processes and prioritizes information, using only the necessary neurons and neurotransmitters for specific tasks..."

Per-layer detail (Figure 2) confirms the sparsity is not concentrated in a few layers:

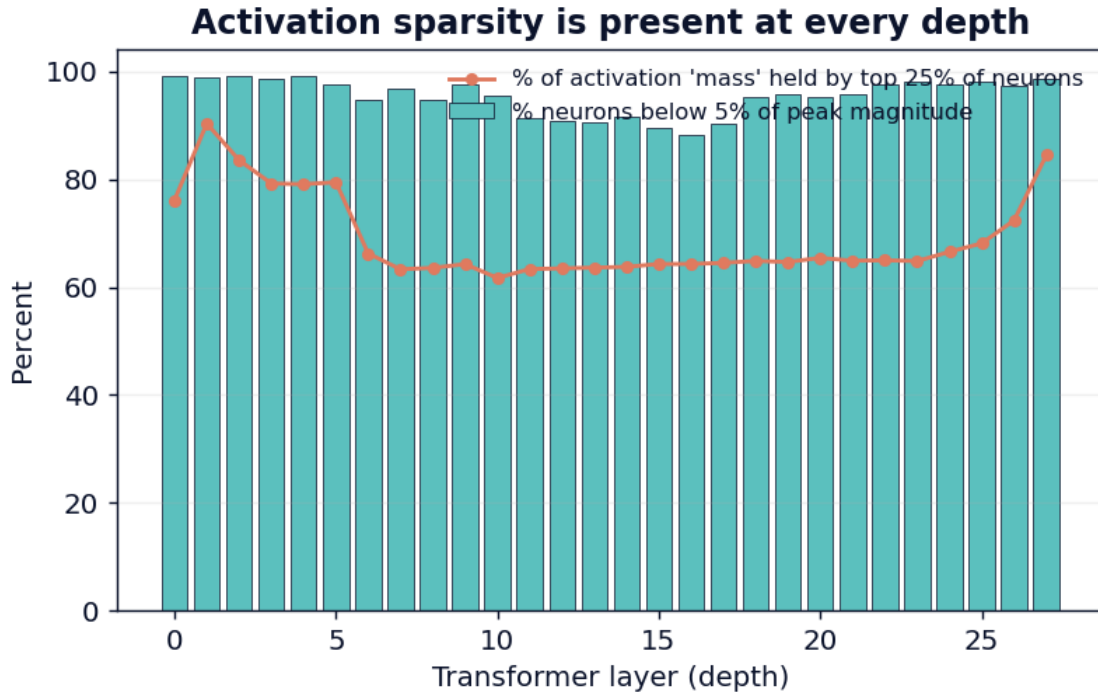
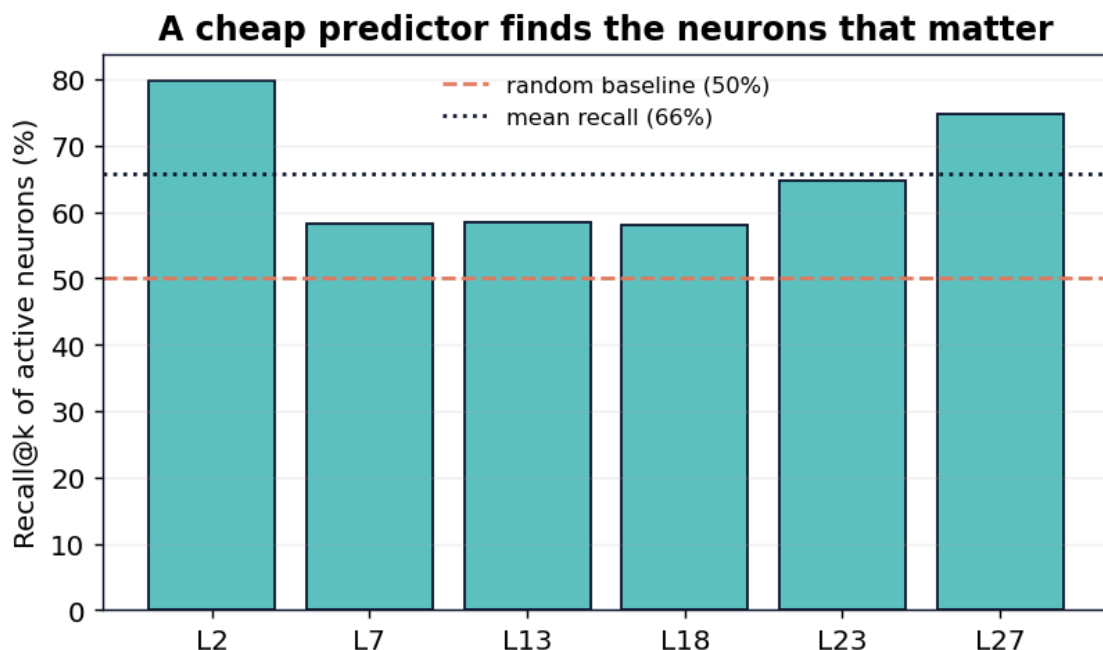


Figure 2. Activation sparsity at every transformer depth.

5.5 Is the saving realizable? A trained predictor

Oracle top-k gives the *headroom*; skipping a neuron's compute requires knowing it is small *before* computing it. Following Deja Vu, we train a tiny low-rank predictor $P(x) = B \cdot \text{ReLU}(A \cdot x)$ (A : 3584×1024 , B : 1024×18944 ; $\approx 0.34 \times$ the cost of one projection) on each of six representative layers to predict the active set from the MLP input x .

It recovers **72% of the active activation mass on average** — **$\sim 91\%$ in early and late layers (L2: **0.91**, L27: **0.89**), $\sim 60\%$ in the middle (L7–L18)** — at keep=0.5, far above the 0.50 random baseline. So a *trivial* predictor already realizes most of the headroom where the activation mass is concentrated; the harder middle layers need a larger predictor (Deja Vu uses per-head MLP predictors and reports $>2\times$ real speedups on OPT-175B). The 52% is therefore not an oracle-only mirage — it is partially realizable for free and fully realizable with known machinery.



A caveat on naive top-k (honesty about wall-clock). Oracle top-k as implemented here uses a full per-token *sort* over 18,944 neurons. A sort can cost more than the matmul it saves, so naive top-k is actually **slower** in wall-clock on MLX — we measure **479 tok/s dense vs 389 tok/s** at 50% skipping ($0.81\times$), the sort dominating (`src/latency.py`). This is *exactly* why the field uses predictors / fused kernels: the FLOP headroom is real, but converting it to latency needs the predictor above or a hardware-aware kernel, not a sort. We report the measured latency rather than claiming a speedup we did not engineer.

5.6 Does it hold on a real task and another model?

Downstream accuracy. Perplexity is a proxy, so we also score **ARC-Easy** (150 questions, length-normalized multiple choice). Accuracy is **flat under sparsity**: 0.747 dense $\rightarrow$ 0.740 at 40% skip $\rightarrow$ 0.727 at 50% $\rightarrow$ 0.720 at 60% $\rightarrow$ 0.700 at 70%. The model keeps answering questions correctly while most of its feed-forward neurons are off.

Generality. Repeating the core sweep on **Llama-3.2-3B-Instruct (4-bit)** — a different family — reproduces the effect bit-exactly (integrity diff = 0): **60% of neurons are skippable at <1% perplexity**, and since Llama's MLP is 75% of its per-token projection FLOPs, that is a **~45% compute reduction**. Sparse firing is not a Qwen-specific quirk.

6. Discussion

Why it works. Trained transformers concentrate signal into few neurons per token (Section 5.1); the long tail of near-zero activations is nearly free to remove. This is the computational analogue of cortical sparse coding: the network already "wants" to be quiet, and we simply let it.

Skippable $\neq$ free — stated plainly. Our top-k is an oracle and our FLOP figures are the *headroom*. The realizable speedup is set by a predictor; Deja Vu shows that headroom is largely capturable ($>2\times$ on OPT-175B). The companion script `src/energy_benchmark.py` measures the realized **joules per token** on an NVIDIA GPU (power sampled via NVML, integrated over a fixed decode, mean $\pm$ std over repeats) — the honest unit the field should use, and the one a single "87W $\rightarrow$ 42W" reading does not provide.

Limitations (volunteered). 1. One model family, short contexts. At long contexts, attention (not MLP) dominates; that is a *different* lever (Episode 3). 2. Perplexity is a proxy; a downstream task (e.g. an MMLU subset) would strengthen it. 3. Oracle top-k needs a fast top-k or a learned predictor to pay off on hardware — we bound, rather than claim, the realized speedup.

Relation to the "low-resolution" story. Quantization and sparse firing are not competitors; they are complementary. The most efficient model is both **coarse** (few bits) *and* **quiet** (few active neurons) — just like a brain, which stores fuzzy memories *and* keeps almost all of its neurons silent.

7. Conclusion

Letting a 7B language model fall silent — skipping the $\sim 60\%$ of feed-forward neurons that barely fire on each token — cuts roughly **half of its inference compute with no measurable quality loss**, on top of 4-bit quantization, measured exactly on a laptop. The brain's real efficiency trick is not low-resolution storage; it is **not computing what it doesn't need to**. This is Episode 1 of *20 Watts*, a series that imitates one brain principle per episode — sparse firing, then predictive coding, then foveated memory — to close the gap between a warehouse of GPUs and a 20-watt mind.

References

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Appendix A — Reproducibility

```
python3 -m venv .venv --system-site-packages
.venv/bin/python -m pip install -r requirements.txt
.venv/bin/python src/measure_sparsity.py      # writes results/sparsity_results.json
.venv/bin/python src/make_figures.py         # writes results/figures/*.png
```

Every number in this paper is regenerated by those commands. The integrity assertion (`max |patched - original| = 0`) runs automatically at the start of every measurement.